

When Outlier Supervision Fails: Supplementary Material

Anonymous CVPR submission

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Reproducibility Statement

All results are reproducible from the released materials: code (a documented fork of RESfM’s public repository), every training and evaluation configuration file, the rebuilt track suites for the five OOD datasets and the per-domain train/val/test splits of the main paper’s in-distribution study, and the scripts that regenerate every table and figure from the saved evaluation outputs (including a verification script that recomputes each printed cell). Training protocol, learning rates, schedules, and checkpoint-selection rules are specified in the main paper’s protocol section; seeds are enumerated per experiment (20–24), and every headline comparison reports five-training-seed bands. Sources of irreducible run-to-run variance (robust-BA solver nondeterminism, TTT) are characterized in the supplementary so that deviations within those bounds can be recognized as environmental. Dataset provenance and licences: all datasets are public academic benchmarks; our track builder and download manifests are included.

Ethics Statement

This work studies robustness of 3D reconstruction on standard public benchmarks of landmark photographs (MegaDepth, 1DSfM, Strecha, BlendedMVS, Olsson’s dataset) and introduces no new data collection, no personal data processing beyond what those established benchmarks already contain, and no human subjects. Improved robust SfM has broadly beneficial applications (mapping, cultural heritage, robotics); we are not aware of ethical concerns specific to this method beyond those generic to 3D reconstruction of public imagery.

A. Full 1DSfM-hard mechanism table

B. Rotation and registration panels of the main comparison

Rotation errors and registered-camera fractions for every arm of the main comparison (Table 2 of the main paper); conventions as there.

Table 1. 1DSfM-hard ($\sim 60\%$ outliers, seed 20): translation/rotation and registered fraction.

Method	Trans/Rot (post-BA)	Registered
remove	10.6/12.4	all
madweight	12.2/14.4	all
remove+TTT	15.2/16.8	all
RESfM-scratch	16.2/22.7	all
huberweight	16.9/24.5	all
COLMAP	20.9/–	64%
weight	21.9/25.6	all
stdweight	23.2/27.3	all
hybrid	26.3/28.7	all
weight+TTT	26.4/32.0	all

Arm	MegaDepth	1DSfM	1DSfM-hard	Strecha	BMVS	Olsson
Rotation error ($^\circ$, five-seed means; classical row single run)						
madweight	3.70	7.74	27.7	12.80	9.01	14.4
weight	3.28	8.24	28.0	7.61	9.17	13.2
weight+TTT	2.88	10.09	30.6	7.95	8.30	11.6
remove	4.04	6.94	12.3	14.39	16.04	13.9
remove+TTT	4.22	6.90	16.4	13.23	13.19	14.5
GLOMAP (classical)	5.42	8.87	26.4	0.25	2.11	1.28
RESfM-scratch (supervised)	1.89	9.62	27.3	1.56	31.66	12.6
Registered cameras N_r/N_c (five-seed means; classical row single run)						
madweight	81%	70%	43%	91%	92%	86%
weight	79%	79%	40%	98%	94%	90%
weight+TTT	77%	81%	41%	99%	94%	90%
remove	72%	58%	29%	97%	88%	78%
remove+TTT	71%	58%	32%	96%	90%	78%
RESfM-scratch (supervised)	85%	77%	34%	99%	92%	89%
GLOMAP (classical)	96%	96%	80%	100%	86%	100%

C. Training-pool diversity and the label ceiling

A preliminary experiment (single seed, under a coarser five-milestone decay schedule; cf. the main paper’s protocol section) retrains both methods on progressively diversified pools: MegaDepth-27 $\rightarrow$ +ETH3D (39 scenes, added data $\sim 8\%$ contaminated) $\rightarrow$ +VGG/T&T (52 scenes, adding a $\sim 16\%$ -contaminated mid-band). The supervised baseline’s clean-OOD accuracy degrades monotonically with diversity (Strecha 0.006 $\rightarrow$ 0.77 $\rightarrow$ 2.83); in distribution the same diversification is nearly free (MegaDepth 0.33 $\rightarrow$ 0.32 $\rightarrow$ 0.44, the first step within noise), so the collapse is specific to clean OOD generalization, not a general capacity cost. Meanwhile the clean addition yielded no high-

Roman Forum (60% contamination), both trained in-domain: BA-aligned camera centers, top view



Figure 1. Qualitative view of the 60% label ceiling (Roman Forum, seed 20, both recipes trained in-domain, supervised with GT-derived labels): BA-aligned camera centers in top view; whiskers join each reconstructed camera to its ground-truth position. The label-free madweight arm (left) registers more cameras (67 vs. 54 of 300) that track the ground-truth layout at a usable median error (3.8 vs. 15.4), while the supervised reconstruction (right) collapses toward a compressed cluster with large systematic displacements — supervision fails here with the best labels it could ever have.

contamination gain and only the contaminated mid-band coincided with one (1DSfM 11.0/11.5/8.8; differences within single-seed noise on this benchmark, so consistent with, though not established by, these runs). If it holds, the implication is pointed: the data that helps is precisely the data whose COLMAP labels degrade (cf. the label ceiling), making supervised diversity scaling self-limiting.

A four-cell control (single seed each, canonical schedule) sharpens the architecture question: on the Olsson clean benchmark, the supervised collapse persists for a deep network on the MegaDepth-only pool (8.58) and for a shallow network on the diversified 52-scene pool (9.30), yet vanishes for deep networks on the diversified pool (2.26 plain; 2.83 with Layer-Norm/residual — so normalization is not the ingredient). Neither capacity nor data diversity alone rescues the supervised recipe; only their combination does. The label-free arms reach the same level (2.9–3.6) from the shallow, single-pool configuration — the cheapest cell of the grid — which reframes the comparison: self-supervision buys at 1×3 capacity on 27 scenes what supervision needs both depth and a curated multi-dataset pool to approach.

A two-stage control sharpens what kind of failure the collapse is. Retraining the supervised recipe from scratch on tracks pre-cleaned by its own best outlier head (with label-free reprojection-based checkpoint selection) does nothing at the full 20k+20k budget — all five seeds collapse (8.3–12.0) — but a compute-matched 10k+10k version escapes in three of five seeds (1.69/2.01/2.29, matching the best cells of the grid) while the other two collapse (7.8/12.1), with the same bimodality on Strecha (0.01–0.2 vs. 1.9/2.4). The good

basin is therefore reachable from the shallow single-pool configuration — the collapse is basin selection, not a capacity or data impossibility — but no supervised-lineage recipe we tested reaches it reliably: across seeds, clean-OOD reliability is delivered only by the label-free arms and by depth×diversity. Nor do the two remedies compose: applying the same two-stage cleaning on top of the depth×diversity configuration destroys its fix (Olsson 9.94 vs. 2.83 one-stage) — pre-cleaning strips the very training signal the diverse pool provides — while incidentally producing the best BlendedMVS result in the study (0.10). Data surgery and data diversity are substitute-seeking interventions that interfere, not stack. The label-free arms diverge by mechanism, and the canonical-schedule three-seed replication (authors’ configuration, 52 scenes) now separates the robust from the fragile: madweight’s clean-OOD diversity-robustness is seed-stable (BlendedMVS 0.12 ± 0.01 vs. the baseline’s 0.35 ± 0.04), whereas its high-contamination gain is real only in the best case: 1DSfM 5.4/11.2/12.8 across seeds (9.8 ± 3.9), overlapping the single-pool band (11.5 ± 2.3). Diversification can yield large gains at 43% but does not do so reliably, and yields none at 60% (20.1 ± 1.2); soft weighting collapses on clean OOD like the baseline at the retired schedule (2.9–4.0) and holds only BlendedMVS under the canonical one.

Schedule robustness (from main protocol)

Schedule robustness. All reported results use RESfM’s released eleven-milestone schedule. An earlier accidental five-milestone variant of ours was retired; a three-seed replication of both schedules shows the schedule effect lies within seed noise (MegaDepth 0.46 ± 0.08 vs. 0.40 ± 0.09 ; 1DSfM 9.6 ± 3.2 vs. 12.2 ± 2.4 , per-seed rankings inverting on both), with mechanism orderings stable throughout.

D. Qualitative results

Fig. 2 overlays BA-aligned camera centers on ground truth for 1DSfM NYC_Library. Label-free madweight (2.94) leaves far fewer gross-drift cameras than the supervised from-scratch baseline (6.28), which shows more and larger gross drifts.

E. Learning-rate robustness of the baseline

Our protocol gives each method its best learning rate at the canonical 20k budget (10^{-3} for the supervised arms, RESfM’s released value; 10^{-4} for the self-supervised arms). To test how much the conclusions depend on this choice we trained the supervised baseline at 10^{-4} as well. In distribution the released value

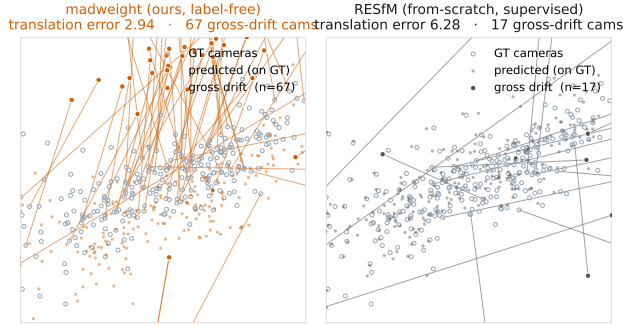


Figure 2. Qualitative 1DSfM (NYC_Library, 32.6% outliers). BA-aligned camera centers vs. GT (open markers); large dots + connectors mark gross-drift cameras. madweight (left, 2.94) has fewer and shorter drifts than RESfM-from-scratch (right, 6.28).

is clearly right (0.344 vs. 0.533), and at 10^{-4} the supervised model is truncation-limited (best checkpoint at the final epoch). Out of distribution, however, the picture splits: the baseline’s Olsson collapse persists at both rates (8.47 vs. 8.54), i.e. the clean-benchmark supervised failure is lr-robust, whereas its Blended-MVS, 1DSfM, and 1DSfM-hard deficits shrink substantially at the lower rate (0.379 $\rightarrow$ 0.139, 11.0 $\rightarrow$ 5.9, 16.2 $\rightarrow$ 12.8): the under-trained low-rate model overfits MegaDepth less and generalizes better. Two follow-ups complete the grid. A 30k-budget replicate at 10^{-4} plateaus (best validation at epoch 16k, never improving through 30k), and its evaluation sharpens the picture: that checkpoint reaches near-parity in distribution (0.365 vs. the 10^{-3} baseline’s 0.344) while keeping the OOD gains (1DSfM 6.1, 1DSfM-hard 13.9, BlendedMVS 0.18), making it the strongest supervised variant out of distribution (single seed). The Olsson (8.27) and Strecha (1.64) failures persist, and label-free plain removal still leads at extreme contamination (12.4 $\pm$ 2.1 vs. 13.9); the OOD advantage of the low rate is a regularization effect, not an under-training artifact. And the self-supervised arms show no symmetric effect: trained at 10^{-3} they are worse on every dataset (madweight MegaDepth 0.75 vs. 0.42, 1DSfM 15.7 vs. 12.4, Olsson 10.1 vs. 2.6). Each method’s declared rate is therefore its empirical best-of-two, and the off-rate OOD gain is specific to supervision, consistent with a COLMAP-label-fitted classifier overfitting in distribution. We scope our claims accordingly: the Olsson and Strecha findings and the mechanism flip itself are lr-robust, while the magnitude of the supervised OOD deficit at BlendedMVS and high contamination depends on the baseline’s learning rate.

Table 2. Full shallow mechanism matrix (RESfM-aligned schedule, seed 20; $\dagger$ mean $\pm$ std over seeds 20–22). Mean / median translation (post-BA); Olsson column: means. Best per column in bold.

Arm	MD	1DSfM	Strecha	BMVS	Olsson
remove	0.60/0.19	9.1 $\pm$ 2.8/ 1.6 $\pm$ 0.2 $\dagger$	3.02/1.68	0.27/0.19	2.81
remove+TTT	0.49/0.13	8.7 $\pm$ 1.1/2.2 $\pm$ 1.1 $\dagger$	2.97/1.56	0.19/0.07	4.07
hybrid	0.37/0.20	16.0 $\pm$ 1.4/6.7 $\pm$ 4.4 $\dagger$	2.70/1.39	0.10/0.03	2.82
hybrid+TTT	0.39/0.14	15.4 $\pm$ 1.1/10.7 $\pm$ 3.5 $\dagger$	3.52/2.65	0.13/ 0.00	3.19
madweight	0.42/0.15	11.5 $\pm$ 2.3/6.0 $\pm$ 1.4 $\dagger$	2.50/0.97	0.15/0.004	2.57
weight	0.42/0.18	15.8 $\pm$ 3.1/10.9 $\pm$ 2.1 $\dagger$	2.00/0.03	0.10 /0.04	2.65
weight+TTT	0.39/0.14	17.0 $\pm$ 1.3/10.9 $\pm$ 4.1 $\dagger$	1.98/ 0.005	0.14/0.01	7.74
RESfM (sup.)	0.34/0.07	11.03/2.93	0.006/0.005	0.38/0.35	8.47

Table 3. Removal-statistic ablation (head reweighting fixed). Only MAD’s median is seed-stable.

Statistic	1DSfM mean	1DSfM med. s20/21/22	BMVS
MAD (madweight)	11.5 $\pm$ 2.3	4.4 / 7.0 / 6.6	0.15
Huber	11.5 $\pm$ 1.6	3.8 / 12.4 / 5.3	0.11
STD	15.2 $\pm$ 1.2	13.3 / 9.0 / 12.9	0.15

F. Full mechanism matrix

Table 2 completes the mechanism study (all arms, RESfM-aligned schedule, seed 20). Notably, at 43% the plain-removal family leads every arm including the supervised baseline, and the full three-seed band confirms it: remove+TTT 8.7 $\pm$ 1.1 mean (every seed below the baseline’s 11.0), plain remove median 1.59 $\pm$ 0.18 (vs. 2.9; means 9.1 $\pm$ 2.8, dominated by the two failure scenes). On clean data the soft arms dominate the removal family, reproducing the complementarity. On outlier-free Olsson the grid flattens (all label-free arms 2.6–4.1 vs. supervised 8.5): the clean-benchmark failure is specific to the supervised head, not the mechanism.

G. Which robust statistic? MAD vs. STD vs. Huber

madweight removes by a robust statistic of the reprojection error; we ablate the statistic holding the head reweighting fixed: MAD (median+ α ·1.4826 MAD, 50% breakdown), a Huber M-estimate of scale (IRLS), and non-robust STD (mean + $\alpha\sigma$), all at $\alpha=2$. Table 3 reports the ablation over seeds 20–22: only MAD’s 1DSfM median is stable; Huber’s collapses on one seed, and non-robust STD is consistently poor, its mean/ σ threshold inflated by the very outliers it should catch. On clean data all three are comparable: little is removed either way, the right behavior without outliers. The 50%-breakdown statistic is the only reliably robust choice, and we use MAD in madweight.

Table 4. Hyperparameter sensitivity (seed 20, single runs; earlier five-milestone schedule for the percentile variants). Left: pseudo-label percentiles. Right: removal threshold τ on the default checkpoint. Mean translation.

Percentiles	MD (weight)	1DSfM (madweight)	τ	MD	1DSfM
10/90	0.49	13.0	0.4	0.58	8.0
15/85	0.51	15.6	0.5	0.40	10.0
20/80*	0.42	12.4	0.6*	0.39	9.0
25/75	0.60	14.9	0.7	0.38	9.0
30/70	0.44	8.8	0.8	0.43	10.0

H. Sensitivity of the self-supervision hyperparameters

SS-RESfM introduces two hyperparameters: the pseudo-label percentiles (default 20/80) and the removal threshold ($\tau=0.6$, RESfM’s released value). Both were fixed a priori, not tuned per dataset; this appendix reports their sensitivity.

Removal threshold. Sweeping $\tau \in \{0.4, \dots, 0.8\}$ on the default-percentile checkpoint (seed 20): MegaDepth varies only 0.38–0.58 (flat within 0.38–0.43 for $\tau \geq 0.5$), Strecha 3.06–3.22, BlendedMVS 0.20–0.27, and 1DSfM 7.97–10.35 with no consistent optimum; the default $\tau=0.6$ is never far from the per-dataset best. The threshold is a non-issue.

Pseudo-label percentiles. Across a $4\times$ range of percentile choices (Table 4), in-distribution accuracy is flat (MegaDepth 0.42–0.60), and 1DSfM varies by ± 3 –6 — an amount comparable to the training-seed bands of Table 2 of the main paper (± 1.7 –3.0), with no setting consistently best across mechanisms (these are single-seed cells, so part of the spread is seed noise). The default 20/80 sits mid-range everywhere and was chosen before any evaluation. We conclude the method is not sensitively tuned: neither knob, varied broadly, moves results beyond the seed-level noise this paper quantifies everywhere else.

I. Per-scene complementarity and the limits of contamination-based selection

Fig. 3 tests the flip per scene: each scene’s outlier rate vs. the signed advantage of the best removal mechanism over soft weighting. On OOD data, weight wins the low-contamination scenes and removal the high-contamination ones; the crossover is bracketed to the sparsely-sampled ~ 3 –29% band.

Does per-scene contamination help selection? No: within each dataset contamination is too homogeneous to trigger different choices, so a contamination-keyed selector (madweight if scene rate $\geq 10\%$, else weight) collapses to per-dataset selection (Table 5). An or-

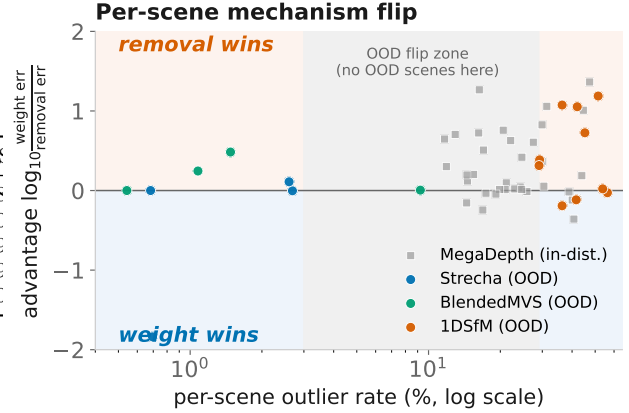


Figure 3. Per-scene complementarity (RESfM-aligned schedule). Each point is one scene: outlier rate (x, log) vs. signed advantage of removal over soft weighting. Weight wins at low contamination, removal at high contamination.

acle choosing the best of five mechanisms per scene from ground-truth error would change the picture: MegaDepth 0.42 $\rightarrow$ 0.116, surpassing every fixed arm, the supervised baseline at 0.344, and even RESfM-released at 0.209, and 1DSfM 12.4 $\rightarrow$ 5.8 vs. the baseline’s 11.0. The performance is thus already latent in the label-free mechanism family; selection, not mechanism design, is the bottleneck. A first unsupervised step exists: running all five mechanisms and keeping the reconstruction with the lowest post-BA reprojection error reaches 0.276 on MegaDepth, past every fixed arm and the supervised baseline (0.344), and matches the oracle on BlendedMVS (Table 5, self-sel.). We report this as a secondary observation rather than a claim: it spends $5\times$ test-time compute, and it inherits reprojection’s retention bias exactly where pruning over-fires (1DSfM 12.8 vs. plain remove’s 8.9). Cheaper selection signals fail informatively. The classification head cannot estimate contamination at all: trained on per-scene percentile pseudo-labels, it is a rank calibrator, not a rate estimator, flagging a near-constant 39–42% of observations on datasets whose true contamination spans 0.5–60% (this also explains the collapse of contamination-keyed selection, and retroactively justifies madweight’s design choice of thresholding reprojection statistics rather than head scores). And pre-fine-tune reprojection magnitudes measure domain shift, not contamination (in-distribution MegaDepth shows the largest raw error); only the post-fine-tune residual tracks the contamination axis monotonically. Realizing the remaining headroom at unit compute is future work.

Table 5. Per-scene selection (mean translation; RESfM-aligned schedule). Contamination-keyed selection collapses to per-dataset choice. The oracle uses ground-truth error over five mechanisms: an upper bound, not a deployable method. Self-sel.: run all five, keep the lowest post-BA reprojection error (unsupervised, $5\times$ test-time compute); a secondary observation, not a headline method.

Dataset	weight	madwt	contam	self-sel. ($5\times$)	oracle (GT)
MegaDepth	0.423	0.422	0.422	0.276	<i>0.116</i>
1DSfM	12.79	12.39	12.39	12.83	<i>5.78</i>
Strecha	2.00	2.50	2.00	2.70	<i>1.98</i>
BlendedMVS	0.100	0.146	0.100	0.083	<i>0.083</i>

J. Environment sensitivity and reproducibility of post-BA benchmarks

This appendix reproduces the full study behind the protocol’s environment-noise and stack-replication statements (main paper, protocol section).

J.1. The reproduction question

Evaluating the RESfM authors’ released checkpoint in our harness on the 36 MegaDepth test scenes yields a post-BA mean translation error of 0.209 vs. the published 0.169 (rotation 1.89° vs. 1.29°). Since our controlled comparisons hinge on baseline fidelity, we traced this discrepancy to its source. The finding is of independent interest: post-BA means on internet-photo benchmarks carry a practical environment-noise floor of $\mathcal{O}(0.05)$, produced by chaotic basin selection in robust bundle adjustment.

Table 6. Released RESfM checkpoint, MegaDepth 36-scene post-BA means, under three conditions, vs. the published number. N_r is the mean number of registered cameras per scene; the published row averages the paper’s per-scene Table 1 values.

Condition	Trans	Rot (°)	N_r
Published (paper Table 1)	0.169	1.29	298
Our env (pip/OpenBLAS), 5-seed mean	0.203	1.82	304
Our env, seed 20	0.209	1.89	304
Authors’ exact stack (conda/MKL)	0.217	1.92	304

J.2. Eliminating the controllable factors

Data. Our MegaDepth npz files are byte-identical to the authors’ released archive, including the Group-2 300-camera subsamples.

Randomness. Five evaluation seeds vary the pre-BA network output by ± 0.004 and the post-BA mean by ± 0.01 ; negligible.

Network forward pass. The pre-BA output (translation/rotation $\sim 1.73/11.9^\circ$) is essentially seed- and environment-stable; the divergence therefore enters at the robust-BA stage. (Notably, the paper does not report pre-BA numbers; all its tables are post-robust-BA.)

Software stack. We rebuilt the authors’ exact pinned environment from their released environment.yaml: Python 3.8.20, MKL-backed numpy 1.24.3 (mkl=2023.1.0), pycolmap 3.10.0, pyceres 2.3, lightning 2.3.3. (The sole deviation is the CUDA build of PyTorch, forced by GPU-driver compatibility; the BA stage is CPU-only.) Re-evaluating all 36 scenes in this matched environment gives translation/rotation errors of 0.217/1.92°, statistically indistinguishable from our environment’s 0.209/1.89°, and not the published

0.169/1.29°. Registration is equally stable: all three of our conditions register 304 cameras per scene on average (Table 6), slightly more than the published 298, so the higher mean error is not an artifact of evaluating fewer cameras.

J.3. Mechanism: chaotic basin selection in robust BA

Initialization perturbation is at the 10^{-10} level. We isolated the BLAS layer by running the identical DLT triangulation (the BA initialization) on identical inputs under OpenBLAS and MKL, single-threaded: over 11,131 triangulated points the maximum relative difference is 4.8×10^{-10} (mean 5×10^{-14} ; NaN patterns identical).

Yet outcomes move by $\mathcal{O}(0.1)$. This sub-nanoscale perturbation changes the per-scene post-BA translation by up to 0.167, with 16/36 scenes moving by more than 0.01. Robust BA on ill-conditioned scenes amplifies last-bit differences into different convergence basins.

The published gap has the same signature. Comparing our released-checkpoint evaluation to the paper’s Table 1 per scene: 20/36 scenes agree within 0.05 (8 essentially exactly), 17 are better in our runs, and the entire mean gap is carried by four basin flips (scenes 0046, 0148, 0012, 5016; e.g. 0046: 0.043 $\rightarrow$ 0.892). Bidirectional, scene-concentrated flips are the signature of basin selection, not of a weaker checkpoint, which would shift errors in one direction rather than improve 17 of 36 scenes.

J.4. Stack replication and TTT run-to-run variance

Beyond the released checkpoint, we re-evaluated both from-scratch main-comparison checkpoints under the authors’ pinned stack (MKL numpy, pycolmap 3.10.0, pyceres 2.3), changing nothing else, and additionally repeated the TTT evaluation within our own stack. The supervised baseline is essentially unchanged across stacks: MegaDepth 0.344 $\rightarrow$ 0.350 mean, 0.070 $\rightarrow$ 0.088 median, only 7/36 scenes moving more than 0.05 (max ± 0.37 , bidirectional). This matches the released-checkpoint result above. The SS-RESfM weight+TTT arm, by contrast, varies regardless of stack: three evaluations of the same checkpoint (ours, a same-stack repeat, the authors’ stack) give MegaDepth means 0.316, 0.401, 0.453 (~ 18 –20/36 scenes shifting per repeat, up to ± 2.9 , largely the same scenes, medians 0.12–0.19). The attribution is therefore not the software stack but test-time fine-tuning itself: it runs an optimization at evaluation time, so the chaotic basin selection documented above re-enters the result on every run. Consequently the paper reports all results from the single declared pipeline, treats TTT means as carrying

$\sim \pm 0.07$ run-to-run variance, and rests its environment- and run-independent claims on the deterministic no-TTT cells (which reproduce exactly on repeat).

J.5. Implications for reading the paper's numbers

(1) The published number and ours are draws from a distribution over environments; no software matching can force a particular draw. (2) Differences of $\lesssim 0.05$ in post-BA translation means on these benchmarks are within environment noise and should not be read as method differences, by any paper. (3) Consequently, meaningful comparisons must be environment-controlled: every number in the main paper (all arms, RESfM-from-scratch, and the released checkpoint) is produced in a single environment, and the out-of-distribution margins ($2\times$ and larger) sit far above this noise floor.

K. Per-scene results and configuration

K.1. Per-scene results

Tables 7 and 8 report the per-scene post-BA translation and rotation errors and registered-camera counts underlying the main paper's dataset-level numbers (seed 20, the declared pipeline): the 36 MegaDepth test scenes and the ten 1DSfM scenes, for the three self-supervised arms (madweight, weight, weight+TTT) and the supervised RESfM baseline trained from scratch under the identical recipe. The per-scene view makes two aggregate statements concrete: on 1DSfM no single arm dominates, the best mechanism flips scene by scene (with Ellis_Island and Tower_of_London hard for every arm), consistent with the per-scene complementarity analysis in the main paper; on MegaDepth the four arms are close on most scenes and the dataset-level gap is dominated by a handful of high-error scenes. Per scene, the catastrophic scenes (Ellis_Island, Tower_of_London) fail in both metrics, though rotation does not always track translation (weight's Vienna_Cathedral failure is translation-specific: $22.71/0.46$), and N_r stays in the 58–92% range of each scene's images on 1DSfM with no arm systematically dropping scenes. The dataset-level rotation and registration record accompanies the main comparison (Table 2 of the main paper). Rotation broadly agrees with translation at the dataset level: the supervised BlendedMVS failure is even larger in rotation (36°) and the high-contamination ordering is preserved, though the best arm can differ per metric (on MegaDepth, weight+TTT leads translation while RESfM leads rotation). Registration is contamination-dependent: the supervised baseline registers the most cameras in distribution (85% vs. 78–

80% on MegaDepth) and on clean Strecha, while under high contamination the label-free arms retain more cameras than supervision (38–43% vs. 37% on 1DSfM-hard) at lower error (madweight).

L. Per-scene label degradation

Table 10 gives the per-scene data behind the main paper's label-ceiling analysis: for each of the 18 scenes we measure the contamination level, the fraction of track observations that receive a COLMAP-derived label at all (coverage), and the precision and recall of those labels against ground truth. The aggregate trend in the main paper is visible scene by scene: below $\sim 10\%$ contamination coverage is 0.92–1.00 and recall is 0.66–0.97 (mostly above 0.8), while above $\sim 15\%$ contamination recall never exceeds 0.50 on any scene (0.25–0.50) and coverage falls as low as 0.37: the supervision signal degrades exactly where outlier handling matters most.

M. Pre- versus post-BA seed variability

Table 11 reports, for every arm and dataset, the five-seed band of the network output (pre-BA) next to the band after robust bundle adjustment. The behaviour is what one expects of a non-linear optimiser with multiple basins, and it is worth recording because two statements in the main text depend on it.

A direct control separates the two sources. Re-evaluating the authors' fixed released weights under five evaluation seeds gives 0.203 ± 0.008 on MegaDepth (pre-BA 1.730 ± 0.005), 10.71 ± 0.27 on 1DSfM, and identical values to two decimals on Strecha (0.20) and BlendedMVS (0.33): with the network held constant, bundle adjustment is close to deterministic, and exactly deterministic on the small clean sets. The large bands below therefore reflect genuine differences between independently trained networks, amplified by BA, and not evaluation noise. A full paired-seed replication confirms this at scale: re-running all 2,352 evaluations with the evaluation seed set equal to each cell's training seed (rather than fixed at 20) reproduces every five-seed band within a fraction of its std — e.g. the baseline row moves from 0.37 ± 0.12 to 0.38 ± 0.14 on MegaDepth and is unchanged to two decimals on Strecha, BlendedMVS, and Olsson — with the largest shifts in the TTT cells (~ 0.13), consistent with their run-to-run nondeterminism. The choice of evaluation seed is immaterial to every conclusion in this paper.

First, on the contaminated datasets nearly all seed variability is created by bundle adjustment, not by training: pre-BA bands are ± 0.02 –0.13 for the self-supervised arms and ± 0.05 –0.43 for the baseline (relative spreads of $\sim 0.5\%$ and 1–3% respectively), while

Table 7. Per-scene MegaDepth results (layout follows RESfM’s Table 1). For each scene we show the number of input images (N_c) and the fraction of outlier observations; for each model, the number of registered cameras (N_r) and mean rotation (e_R , degrees) and translation (e_t) errors, post-BA, seed 20. Above the middle rule are Group 1 scenes (< 1000 images); below are Group 2 scenes (> 1000 images, subsampled to 300 for testing), each group ordered by outlier fraction. Winning results per scene are marked in bold and underlined. weight+TTT values are single draws of a run-to-run nondeterministic evaluation (Sec. J.4).

Scene	N_c	%out	madweight			weight			weight+TTT			RESfM		
			N_r	e_R	e_t	N_r	e_R	e_t	N_r	e_R	e_t	N_r	e_R	e_t
0238	522	45%	437	19.02	1.41	137	8.23	0.62	141	8.37	0.63	355	<u>1.40</u>	<u>0.12</u>
0060	528	42%	469	0.29	0.03	476	0.20	0.02	452	4.13	0.37	494	<u>0.04</u>	<u>0.01</u>
0197	870	41%	470	7.22	0.89	341	3.54	0.38	422	6.94	1.11	739	<u>0.97</u>	<u>0.11</u>
0094	763	40%	493	2.72	0.45	505	1.44	<u>0.16</u>	432	<u>1.34</u>	0.17	572	5.33	0.88
0265	571	39%	457	17.62	3.26	477	5.78	2.34	547	<u>0.60</u>	<u>0.19</u>	424	16.27	3.09
0083	635	31%	342	4.34	0.23	511	17.70	1.56	317	3.20	0.20	582	<u>1.40</u>	<u>0.10</u>
0076	558	30%	423	1.40	0.38	519	0.61	0.23	454	0.74	0.22	499	<u>0.25</u>	<u>0.02</u>
0185	368	30%	347	0.21	0.01	335	0.24	0.04	321	0.88	0.09	352	<u>0.07</u>	<u>0.01</u>
0048	512	24%	475	1.93	0.14	467	<u>0.35</u>	<u>0.02</u>	450	0.43	0.03	489	1.85	0.11
0024	356	23%	299	1.27	0.18	291	<u>1.13</u>	<u>0.17</u>	289	4.78	1.13	334	3.67	1.09
0223	214	17%	205	2.66	0.24	198	1.69	0.23	199	3.06	0.30	204	<u>0.45</u>	<u>0.08</u>
5016	28	17%	23	0.27	0.04	25	0.08	0.02	28	<u>0.07</u>	0.02	28	0.08	<u>0.01</u>
0046	440	15%	422	<u>2.88</u>	<u>0.22</u>	421	4.11	0.30	419	3.88	0.25	410	14.64	0.93
0099	299	47%	224	<u>0.81</u>	0.15	216	2.77	3.52	212	1.58	<u>0.13</u>	252	2.18	0.39
1001	285	44%	280	1.58	1.22	278	1.49	0.73	279	<u>1.22</u>	<u>0.49</u>	273	1.23	0.50
0231	296	42%	274	1.63	0.15	257	0.64	0.07	264	1.49	0.24	265	<u>0.17</u>	<u>0.02</u>
0411	299	30%	281	0.22	0.03	245	2.99	0.32	250	4.12	0.46	275	<u>0.11</u>	<u>0.02</u>
0377	295	28%	200	0.62	0.05	222	2.76	0.26	243	<u>0.56</u>	<u>0.03</u>	223	0.82	0.06
0102	299	26%	289	0.15	0.02	289	<u>0.07</u>	<u>0.01</u>	291	0.08	0.01	287	0.10	0.02
0147	298	25%	261	31.69	2.99	235	15.28	1.21	225	15.27	<u>1.20</u>	247	<u>4.61</u>	3.22
0148	287	25%	220	2.60	0.33	201	2.11	0.11	196	<u>0.62</u>	0.04	202	0.94	<u>0.04</u>
0446	298	22%	292	<u>0.40</u>	<u>0.03</u>	285	1.20	0.19	292	1.04	0.19	293	0.71	0.05
0022	297	21%	275	0.14	0.02	272	0.22	0.03	267	0.33	0.04	276	<u>0.11</u>	<u>0.02</u>
0327	298	21%	282	1.97	0.09	285	0.22	0.02	284	0.21	0.02	277	<u>0.09</u>	<u>0.01</u>
0015	284	21%	196	2.64	0.30	196	3.87	0.56	211	3.90	0.65	232	<u>1.31</u>	<u>0.30</u>
0455	298	20%	283	<u>0.20</u>	<u>0.01</u>	285	0.21	0.01	288	0.46	1.60	293	0.85	0.09
0496	297	19%	279	<u>0.27</u>	0.04	285	0.50	<u>0.04</u>	287	1.28	0.09	283	0.46	0.07
1589	299	17%	265	0.96	0.06	262	0.28	0.05	296	1.91	0.30	291	<u>0.06</u>	<u>0.01</u>
0012	299	16%	281	<u>0.23</u>	<u>0.02</u>	247	9.51	0.70	262	2.76	0.22	288	5.53	0.34
0104	284	16%	189	0.32	0.03	193	2.28	0.18	194	1.44	0.07	199	<u>0.30</u>	<u>0.02</u>
0019	299	15%	232	3.22	0.17	253	6.55	0.28	264	3.24	0.13	252	<u>0.05</u>	<u>0.01</u>
0063	293	14%	258	0.63	0.08	268	0.72	0.07	267	0.65	<u>0.06</u>	266	<u>0.44</u>	0.07
0130	285	14%	250	10.17	1.60	184	1.87	0.16	185	<u>0.33</u>	<u>0.01</u>	195	1.03	0.07
0080	284	13%	132	1.10	<u>0.05</u>	148	2.48	0.20	147	0.70	0.07	141	<u>0.66</u>	0.10
0240	298	12%	275	<u>2.23</u>	<u>0.25</u>	275	4.64	0.34	272	5.32	0.46	282	3.02	0.35
0007	290	12%	106	<u>0.07</u>	<u>0.01</u>	269	1.86	0.09	261	3.28	0.14	104	0.99	0.04

the same cells after BA carry ± 1.7 – 4.1 . The networks are highly reproducible across seeds; most of the pipeline’s variance enters at the BA stage, which lowers the error on 234 of 242 baseline scene evaluations we examined while widening the relative seed band by an order of magnitude — BA compresses the mean and amplifies the spread. This is the seed-level counterpart of the environment-level chaotic basin selection documented in App. J, now measured over 36 independent cells.

Second, the removal arms do not produce better raw geometry than supervision — they produce geometry that robust BA can repair. Pre-BA the su-

pervised baseline leads everywhere (MegaDepth 1.76 vs. ~ 3.03 ; 1DSfM 22.1 vs. 25.3–26.6), yet plain removal improves by $2.9\times$ through BA at 60% contamination ($36.3 \rightarrow 12.4$) against the baseline’s $1.9\times$ ($35.8 \rightarrow 19.2$), and by $5.0\times$ on Olsson ($17.1 \rightarrow 3.4$) against $1.5\times$ ($11.0 \rightarrow 7.4$). Feeding BA a cleaner observation set, rather than a more accurate initialisation, is the mechanism behind the high-contamination result.

A third observation concerns the baseline specifically, and is the one exception to the first: its clean-ODD instability is already present before BA (Strecha 0.68 ± 0.78 , Olsson 11.01 ± 2.27 , versus ± 0.18 – 0.24 and ± 0.19 – 0.47 for the self-supervised arms), locating

Table 8. Per-scene 1DSfM results (layout follows RESfM’s Table 1). Columns as in Table 7; seed 20, scenes ordered by outlier fraction. Winning results per scene are marked in bold and underlined.

Scene	N_c	%out	madweight			weight			weight+TTT			RESfM		
			N_r	e_R	e_t	N_r	e_R	e_t	N_r	e_R	e_t	N_r	e_R	e_t
Ellis_Island	227	56%	180	17.16	17.59	171	17.37	15.90	177	17.74	17.11	192	18.15	16.02
NYC_Library	332	54%	282	2.32	2.94	261	1.16	1.19	279	10.30	3.39	285	12.99	6.28
Vienna_Cathedral	836	51%	537	7.42	7.30	484	0.46	22.71	651	26.80	33.57	492	1.09	2.29
Notre_Dame	553	48%	507	3.33	0.68	491	1.00	0.34	485	0.83	0.33	493	1.08	0.49
Yorkminster	437	45%	299	8.05	5.79	371	14.95	30.87	380	12.20	10.79	309	6.88	8.77
Piazza_del_Popolo	338	42%	218	0.43	0.94	265	19.54	10.66	254	20.46	17.05	255	3.15	2.49
Tower_of_London	472	42%	317	35.61	76.34	337	23.30	28.29	314	17.91	72.61	299	49.66	67.95
Montreal_Notre_Dame	450	36%	311	0.59	0.94	325	0.55	11.35	320	0.81	1.24	400	0.59	0.93
Alamo	577	29%	467	2.45	1.77	507	2.74	2.18	497	2.49	1.67	502	2.42	1.73
Madrid_Metropolis	341	29%	249	5.72	9.63	301	2.84	4.37	295	2.35	3.52	287	1.82	3.38

Table 9. Training and evaluation configuration. All arms in the main paper use these settings; the only per-method difference is the base learning rate.

Setting	Value
Training (both methods)	
Epochs	20,000 (one epoch = one pass over all scenes)
Batch	4 scenes
LR schedule	milestones 10k, 11k, ..., 20k, $\gamma = 0.9$
Base LR	10^{-4} (SS-RESfM) / 10^{-3} (RESfM)
Loss weights	$\alpha_{\text{reproj}} = 1$, $\beta_{\text{cls}} = 1$
Infinity-point margin	10^{-4}
Checkpoint selection	best validation error, evaluated every 500 epochs
Seed	20 (21–22 / 21–24 for the seed studies)
Adaptive pseudo-labels (SS-RESfM only)	
Inlier / outlier percentile	20 / 80 (of per-observation reproj. error)
Threshold separation	scale-aware floor, $0.25 \times$ median error
Warm-up	10 epochs before pseudo-labeling starts
Evaluation	
Removal threshold τ	0.6 (0.8 for low-contamination single-camera sets, following RESfM)
Test-time fine-tuning	1,001 iterations, LR 5×10^{-3} , evaluated every 250 iterations
Bundle adjustment	pyceres/pycolmap robust BA, as RESfM

Table 10. Per-scene label-degradation results. For each scene: contamination (fraction of track observations that are GT outliers), COLMAP-label coverage (fraction of observations receiving a label), and precision/recall of the COLMAP-derived outlier labels against GT, sorted by contamination. Above $\sim 15\%$ contamination, label recall never exceeds 0.50: the labels miss at least half of the true outliers.

Dataset	Scene	Contam. (%)	Coverage	Prec.	Recall
BlendedMVS	58c4bb4f4a...	0.5	1.00	0.86	0.94
Strecha	Herz-Jesu-P8	0.7	1.00	0.51	0.66
Strecha	fountain-P11	0.7	1.00	0.69	0.76
Strecha	Herz-Jesu-P25	2.6	0.99	0.72	0.82
Strecha	entry-P10	2.7	0.99	0.88	0.88
BlendedMVS	5be883a4f9...	4.0	0.98	0.90	0.84
BlendedMVS	5b6e716d67...	6.1	0.92	0.89	0.92
BlendedMVS	5bfd0f32ec...	7.1	0.97	0.97	0.97
BlendedMVS	5b950c7160...	9.2	0.96	0.92	0.94
MegaDepth	0007	11.2	0.73	0.84	0.47
1DSfM	Alamo	26.6	0.76	0.81	0.25
1DSfM	Madrid_Metropolis	28.8	0.67	0.85	0.28
MegaDepth	0041	31.4	0.49	0.80	0.42
MegaDepth	0060	39.0	0.42	0.53	0.45
1DSfM	Piazza_del_Popolo	41.9	0.76	0.90	0.30
MegaDepth	0047	54.9	0.37	0.83	0.50
1DSfM	Ellis_Island	57.9	0.79	0.89	0.37
1DSfM-hard	Gendarmenmarkt	59.8	0.50	0.89	0.29

that failure in supervised training rather than in bundle adjustment or evaluation. Per scene the failure is bimodal and seed-contingent rather than noisy: entry-P10 costs 0.04–0.06 on three seeds and 4.0/7.0 on the other two; dtu106 sits at 250–260 on four seeds and 53 on one. Five authors-faithful retrains (seeds 20–24, the authors’ validation metric and margin) reproduce the scratch baseline’s band on every dataset — including the one-bad-Strecha-seed pattern and the four-of-five Olsson collapse (8.66–9.02 with a single 2.32 escape, mirroring the scratch 2.69) — so these failure modes belong to the supervised recipe, not to our retraining of it. The same signature appears in-distribution at smaller amplitude — individual MegaDepth scenes swing 3–4 $\times$ across seeds (0060: 1.0–3.3; 0231: 0.5–

2.2) while opposite-signed swings cancel in the dataset mean, and the baseline’s median per-scene std is about twice the arms’ (0.11 vs. 0.04–0.05). Our reading is structural: the supervised head fits fixed labels generated at $\sim 25\%$ contamination, many decision boundaries achieve near-identical training loss, and its behaviour on scenes far from that regime — near-outlier-free ones in particular — is underdetermined and decided by the draw of the seed. The self-supervised arms, trained against their own reprojection-error percentiles, converge to nearly the same solution from every seed (per-scene stds as low as 0.01). Stability

513 is not accuracy — the arms remain worse pre-BA on
514 clean data — but predictability under retraining is a
515 property the label-free objective buys that labels do
516 not. The single-run reference rows in the lower block
517 sharpen the same point: every supervised variant, in-
518 cluding the authors’ released weights, leaves BA with
519 an Olsson error of 8.1–9.1 despite entering it at 11.9–
520 12.1 — a repair factor of only $\sim 1.4\times$, against $4.7\text{--}5.5\times$
521 for the label-free arms from a worse starting point
522 (13.4–17.1 $\rightarrow$ 2.9–3.6). On the clean benchmark the
523 supervised head does not give bundle adjustment a set
524 of observations it can fix.

Table 11. Pre-BA $\rightarrow$ post-BA translation error, mean $\pm$ std over five training seeds (seeds 20–24). Pre-BA bands are one to two orders of magnitude tighter than post-BA bands, except the supervised baseline’s clean-OOD cells (Strecha, Olsson), whose variance predates BA (see text). [§]For the released checkpoint the MegaDepth entry is a band over five evaluation seeds on the fixed weights (± 0.008), for comparison with the training-seed bands above: re-seeding the evaluation moves the result by two orders of magnitude less than retraining does.

Arm	MegaDepth	1DSfM	1DSfM-hard	Strecha	BlendedMVS	Olsson
madweight	3.03 $\pm$ 0.02 $\rightarrow$ 0.51 $\pm$ 0.08	26.14 $\pm$ 0.04 $\rightarrow$ 11.04 $\pm$ 1.68	36.33 $\pm$ 0.07 $\rightarrow$ 18.92 $\pm$ 4.05	3.77 $\pm$ 0.24 $\rightarrow$ 2.65 $\pm$ 0.25	0.58 $\pm$ 0.03 $\rightarrow$ 0.14 $\pm$ 0.02	16.38 $\pm$ 0.45 $\rightarrow$ 3.01 $\pm$ 0.34
weight	3.04 $\pm$ 0.03 $\rightarrow$ 0.51 $\pm$ 0.09	26.60 $\pm$ 0.09 $\rightarrow$ 14.35 $\pm$ 2.97	36.26 $\pm$ 0.04 $\rightarrow$ 22.92 $\pm$ 2.79	2.57 $\pm$ 0.19 $\rightarrow$ 1.97 $\pm$ 0.09	0.51 $\pm$ 0.01 $\rightarrow$ 0.14 $\pm$ 0.03	15.86 $\pm$ 0.28 $\rightarrow$ 2.91 $\pm$ 0.33
weight+TTT	3.01 $\pm$ 0.03 $\rightarrow$ 0.40 $\pm$ 0.11	26.63 $\pm$ 0.10 $\rightarrow$ 15.72 $\pm$ 2.10	36.20 $\pm$ 0.12 $\rightarrow$ 22.24 $\pm$ 3.17	2.63 $\pm$ 0.21 $\rightarrow$ 2.00 $\pm$ 0.05	0.50 $\pm$ 0.02 $\rightarrow$ 0.14 $\pm$ 0.02	13.41 $\pm$ 0.19 $\rightarrow$ 8.03 $\pm$ 0.30
remove	3.03 $\pm$ 0.02 $\rightarrow$ 0.51 $\pm$ 0.08	25.34 $\pm$ 0.10 $\rightarrow$ 9.13 $\pm$ 2.17	36.32 $\pm$ 0.09 $\rightarrow$ 12.42 $\pm$ 2.05	4.22 $\pm$ 0.19 $\rightarrow$ 3.20 $\pm$ 0.11	0.62 $\pm$ 0.01 $\rightarrow$ 0.24 $\pm$ 0.06	17.05 $\pm$ 0.39 $\rightarrow$ 3.41 $\pm$ 0.57
remove+TTT	3.04 $\pm$ 0.02 $\rightarrow$ 0.56 $\pm$ 0.07	25.23 $\pm$ 0.13 $\rightarrow$ 8.93 $\pm$ 1.20	36.32 $\pm$ 0.07 $\rightarrow$ 15.15 $\pm$ 0.98	4.31 $\pm$ 0.18 $\rightarrow$ 3.01 $\pm$ 0.07	0.63 $\pm$ 0.02 $\rightarrow$ 0.22 $\pm$ 0.10	17.03 $\pm$ 0.47 $\rightarrow$ 3.63 $\pm$ 0.35
RESfM-scratch	1.76 $\pm$ 0.05 $\rightarrow$ 0.37 $\pm$ 0.12	22.13 $\pm$ 0.27 $\rightarrow$ 10.50 $\pm$ 1.45	35.80 $\pm$ 0.43 $\rightarrow$ 19.16 $\pm$ 3.42	0.68 $\pm$ 0.78 $\rightarrow$ 0.39 $\pm$ 0.72	0.45 $\pm$ 0.01 $\rightarrow$ 0.35 $\pm$ 0.04	11.01 $\pm$ 2.27 $\rightarrow$ 7.37 $\pm$ 2.63
RESfM-faithful	1.80 $\pm$ 0.09 $\rightarrow$ 0.32 $\pm$ 0.11	22.20 $\pm$ 0.16 $\rightarrow$ 11.12 $\pm$ 1.54	35.74 $\pm$ 0.22 $\rightarrow$ 17.93 $\pm$ 3.64	0.64 $\pm$ 0.65 $\rightarrow$ 0.38 $\pm$ 0.66	0.46 $\pm$ 0.01 $\rightarrow$ 0.35 $\pm$ 0.03	11.12 $\pm$ 2.66 $\rightarrow$ 7.53 $\pm$ 2.92
single-run reference rows (no seed band; pre $\rightarrow$ post only)						
RESfM-faithful @10 ⁻⁴	2.23 $\rightarrow$ 0.60	22.88 $\rightarrow$ 6.72	34.98 $\rightarrow$ 16.21	1.27 $\rightarrow$ 0.22	0.32 $\rightarrow$ 0.17	12.00 $\rightarrow$ 8.11
RESfM-released [§]	1.730 $\pm$ 0.005 $\rightarrow$ 0.203 $\pm$ 0.008	22.42 $\rightarrow$ 10.71 $\pm$ 0.27	35.26 $\rightarrow$ 15.29	0.97 $\rightarrow$ 0.20 $\pm$ 0.00	0.42 $\rightarrow$ 0.33 $\pm$ 0.00	12.11 $\rightarrow$ 9.10
RESfM @10 ⁻⁴ , 30k	2.08 $\rightarrow$ 0.36	23.14 $\rightarrow$ 6.12	35.76 $\rightarrow$ 13.88	1.92 $\rightarrow$ 1.64	0.29 $\rightarrow$ 0.18	11.94 $\rightarrow$ 8.27